Kai Ren

Kyoto University | Ph.D. Candidate | Intelligent Science and Technology
EMG Sensing | Robotics | Human-Computer Interaction
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Profile

Ph.D. candidate in Intelligent Science and Technology at Kyoto University, focusing on electromyography (EMG) sensing, deep learning, and human-computer interaction. Current work targets physiological signal modeling and motion intent prediction for seat-based and exoskeleton assistive systems, with publications in robotics and bioengineering. Experienced in end-to-end workflows from data acquisition and analysis to neural network modeling, real-time communication, and system integration for wearable assistive and embodied-AI applications.

Research Experience

RIKEN (Human-Robot Collaboration Team), Japan *Dec 2021 – Present*
Research Intern
RIKEN-GRP Homepage

- Conduct EMG-based analysis and prediction for sit-to-stand and stand-to-sit transitions in assistive scenarios.
- Design real-time physiological signal processing pipelines and neural network models for motion intent prediction.
- Build and integrate experimental closed-loop systems for powered assistive devices, including seat-based prototypes.
- Support EMG hardware prototyping, including electrode fabrication, soldering, and acquisition-system setup.
- Execute full research workflow: data collection, model training, online evaluation, and system-level validation.

Education

- Kyoto University, Japan, Ph.D. in Intelligent Science and Technology (Apr 2021 – Present)
- Ocean University of China, China, M.Eng. in Computer Application Engineering (2017 – 2020)
- Ocean University of China, China, B.Eng. in Computer Science and Technology (2013 – 2017)

Activities and Projects

- 2014 ACM-ICPC Shandong Provincial Contest participant.
- 2016 Project Lead, SRDP Program, Ocean University of China (iOS app development).
- 2025 IST Best Poster Award, Graduate School of Informatics, Kyoto University.

Scholarships

- Third-Class Academic Scholarship (2017–2020, awarded consecutively each year).

Publications

1. Practical Motion Classification and Forecast toward Safe Assists of Sit-to-stand and Stand-to-sit Motions. *IEEE IROS, 2026*. Under review.
2. Anticipatory prediction of sit-to-stand and stand-to-sit transitions: a unified approach. doi:10.3389/fbioe.2026.1792582. *Frontiers in Bioengineering and Biotechnology*, 2026.
3. Integrated Motion State Prediction for Sit-to-Stand and Stand-to-Sit Motions Toward Effective Power Assist Control. *IEEE ICRA*, 2025, pp. 947–952.
4. Q-DFN: Q-learning Decisive Fusion Network for Object Detection. *Proceedings of CSAI 2019*, pp. 88–92.

Technical Skills

- Programming: Python, MATLAB
- Machine Learning: PyTorch, deep learning for biosignals and time-series modeling
- Signal Processing: EMG acquisition, preprocessing, feature extraction, real-time processing
- Systems & Control: real-time inference pipelines, closed-loop experimental systems, assistive-device integration
- Hardware Prototyping: electrode fabrication, soldering, circuit assembly, EMG sensor setup
- Tools: Linux, Git

Research Interests and Languages

Research Interests: physiological signal modeling for motion intent prediction, multimodal perception with real-time reasoning, and human-centered AI for adaptive interfaces.
Languages: English (TOEFL), Japanese (JLPT N2)